

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Sampling distributions and the central limit theorem

Week 7 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



The point where the course splits into two

Weeks 1-6 — probability. We assumed that the original set was known: μ , σ , ρ were given. From that, we inferred the sample.

Model $\rightarrow$ Data

A common mistake But in real life, it's never like that. We only have one sample. μ is unknown — that's what we want to know.

Data $\rightarrow$ Model

This requires an answer to one question: how reliable is the number calculated from the sample?

Key idea — this is not a simple number. It depends on the sample, and the sample is random. So μ itself is a random variable and has a distribution.



Learning Objectives for Week 7

By the end of this week, you will:

1. Define and list a complete sampling distribution.
2. Prove that $E[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$ (from the formula from Week 6).
3. Calculate the Standard Error and distinguish it from the Standard Deviation.
- 4. Distinguish between the Law of Large Numbers and the Central Limit Theorem.**
5. Examine the conditions for applying the MLT and show when it does not work.
6. Construct a sampling distribution for the proportion $\hat{p}$.

If you were to repeat the study, what number would you get? How different is it? And how do we know?

CLO5 — Explain the role of sampling distributions and MLT in inference. Bloom's Level: Comprehension and Application.

Assessment: Week 8 — Midterm Exam (Weeks 1-7).

Parameters and statistics

Two types of numbers: parameter and statistic

Parameter — a numerical characteristic of a set.

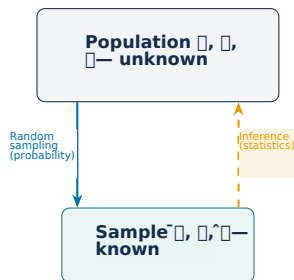
$\mu, \sigma, \rho, \dots$. Greek letters.

Fixed number. Not random. Usually unknown.

Statistics — a number calculated from a sample.

$\bar{x}, s, \hat{\mu}, \dots$. Latin letters.

Random. Different in each sample. Always known (because we can calculate it).



Common mistake

Students often confuse μ and $\bar{x}$, and σ and s . Rule: Greek = population, Latin = sample. No exceptions.

Sampling Distribution Take all possible samples of size n from the original population and calculate the same statistic for each. The distribution of these values is the sampling distribution of this statistic.

A common mistake

Don't confuse the three distributions:

**the original population all n units
sample the distribution of the n units
in your hand the statistics over all
possible samples**

Why is this concept difficult?

Because it is invisible. You only see one sample in your lifetime.

The sampling distribution is a thought experiment. But it can be calculated exactly, and that's what we're going to do now.

In the next slide, we will enumerate all the samples from a real prime set of size $n = 5$ — the thought experiment becomes a real table.

Full enumeration: $n = 5$, $k = 2$

Master set: 2018 wheat yield (t/ha), five Central Asian countries. Source: OWID / FAOSTAT.

Country
Uzbekistan 4,126
Tajikistan 3,048
Kyrgyzstan 2,427
Kazakhstan 1,228
Turkmenistan 1,068

μ 2,3794

σ 1,1442

Conditions

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} = \frac{6,5454}{5} = 1,3091$$

For master set, divide by n — not by $n-1$.

All $(5 \cdot 2) = 10$ samples (without replacement)

Sample 1	Sample 2
Uzbek Kyrgyz 3.2765	Kyrgyz Tajik 2.7375
Uzbek Kazakhstan 2.6770	Kyrgyz Turk 1.7475
Uzbek Tajik 3.5870	Kazakh Tajik 2.1380
Uzbek Turk 2.5970	Kazakh Turk 1.1480
Kyrgyz Kazakhstan 1.8275	Tajik Turk 2.0580

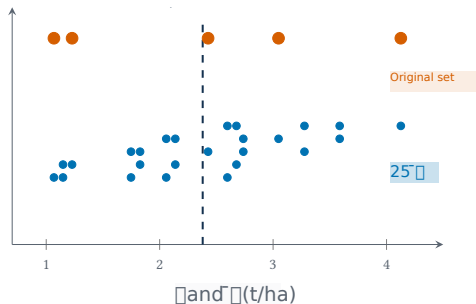
Average of the averages: $3.2765 + \dots + 2.0580$

$$= 2,3794 = \mu.$$

10

It is not a coincidence — we will prove this in the next slide.

Same original set, with resampling: 25 samples



Top row — five actual values. Bottom five rows — all 52 = 25 sample means.

2. The variance has decreased: $SD(\bar{x}) = 0.8090$ vs $s = 1.1442$.

3. The shape has changed: 5 outliers in the original set, 15 in the $\bar{x}$ s, and centered.

Audience Question Where did 0.8090 come from?

$$\frac{s}{\sqrt{n}} = \frac{1.1442}{\sqrt{2}} = 0.8090$$

$E[\cdot]$ and $\text{Var}(\cdot)$

Proof — and it follows from week 6

$\bar{X}_1, \dots, \bar{X}_n$ — independent, uniformly distributed (m.b.b. $\sim$), $E[\bar{X}_i] = \mu$, $\text{Var}(\bar{X}_i) = \sigma^2/n = 1/n$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Center — linearity of expectation (week 4)

$$\begin{aligned} E[\bar{X}] &= E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] \\ &= \frac{1}{n} \cdot n\mu = \mu \end{aligned}$$

Independence is not necessary! Just linearity is enough. $\bar{X}$ — unbiased estimator of μ .

Variance — Formula for Week 6

Week 6: $\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right)$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \right] \\ &= \frac{1}{n^2} [n\sigma^2 + 0] = \frac{\sigma^2}{n} \end{aligned}$$

Common mistake Attention: where is the use of independence? Only in one place — when writing $\text{Cov}(X_i, X_j) = 0$. If the observations are related, this term does not vanish and $\sigma/\sqrt{n}$ is incorrect. Cluster sampling, time series, students in the same class — all have $\text{Cov} > 0$, so the true error is greater than $\sigma/\sqrt{n}$.

Standard Error — and What It IS NOT

Standard Error

$SE(\bar{x}) = SD(\bar{x})/\sqrt{n}$ $\sqrt{n}$ is practically unknown? $SE =$

The Most Common Confusion

n (or N) How much the observations differ. It doesn't change as n increases. SE How much the means differ. It gets smaller as n increases.

When the article says “ 4.5 ± 0.1 ” — what is 0.1: n or SE ? Always ask.

Movie budget: $\bar{x} = \$37.40$ million

n SE vs. $n=5$

5	16,728	1,000
25	7,481	0,447
100	3,740	0,224

Audience question

$\sqrt{n}$ law: to double the precision, you need to quadruple the sample size. And the budget increases by 4x — the returns diminish.

Why $n-1$? — the answer is in 25 samples

In week 1, you asked why σ^2 is divided by $n-1$. Now let's calculate the answer.

From the above $n=5$ sample set, we calculate two different variances for all 25 samples:

Estimator Mean Conclusion		
$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$	1,3091	$= \sigma^2 \checkmark$
$s_n^2 = \frac{\sum (x - \bar{x})^2}{n}$	0,6545	$= \frac{n-1}{n} \sigma^2 \times$
True σ^2 1.3091		

The reason is that $\sum (x_i - \bar{x})^2$ is always less than $\sum (x_i - \mu)^2$ — because $\bar{x}$ is exactly the point that minimizes this sum.

Since we are calculating $\bar{x}$ from the data, one degree of freedom is “wasted”. Dividing by $n-1$ compensates for this loss.

Common Error

Non-stationarity — true for σ^2 , but not for σ : $E[\sigma] \neq \sigma$ (because $\sqrt{\cdot}$ is not linear). Slightly underestimates σ for small n .

Selection without replacement: finite prime set correction

Finite prime set correction (fpc)

$$\text{Var}(\bar{y}) = \frac{1}{n} \cdot \underbrace{\frac{N-n}{N-1}}_{\text{fpc}}$$

$N = 5$, $n = 2$ in our example: $5 - 2$

$$= 0.75, \sqrt{0.75} =$$

0.8660.

Without replacement: SE = 0.7006 With replacement: SE = 0.8090 Ratio: 0.8660

When to consider?

$\alpha / \sqrt{\text{fpc}}$ Decision

0.01	0.995	ignore
0.05	0.975	ignore
0.10	0.949	limit 0.50
0.707		condition

Practical rule: if $\alpha < 0.05$, drop fpc. We will do this for the rest of the course.

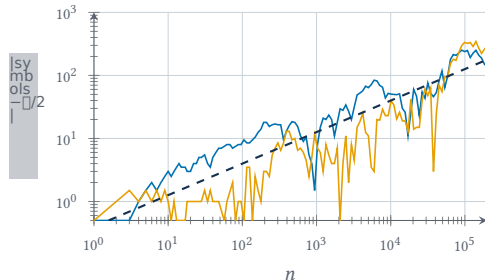
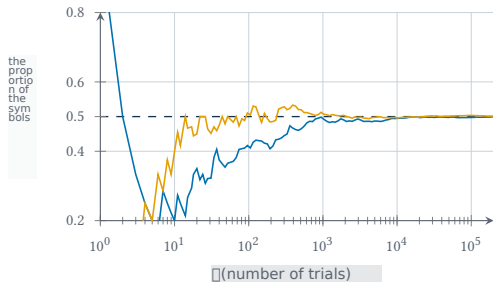
Note: SE does not depend much on the size of the sample — only on n . The same n is needed for both the population of Uzbekistan and the population of Tashkent.

Law of Large Numbers

The Law of Large Numbers — and its Two Faces

The Law of Large Numbers (LNL)

1, 2, ... m.b.b. $\sim$, $E[\bar{X}] = \mu$ is finite, then for any $\epsilon > 0$ $P(|\bar{X} - \mu| > \epsilon) \rightarrow 0$ ($n \rightarrow \infty$): the sample mean approaches the population mean.



Two independent paths. Left: the proportion approaches 0.5. Right: In the same way, the difference between the number of emblems and $n/2$

increases — the dotted line is $0.5\sqrt{2n}$.

"The Law of Averages" — the Most Costly Fallacy

Gambler's Fallacy

"It's been 10 heads in a row — now it must be heads, because there's a law of large numbers."

This is incorrect. A coin has no memory. The KSQ doesn't compensate for past deviations — it dilutes the deviations with new tosses.

Discussion — 3 minutes

The head of the insurance company: "This year we paid a lot. According to the law of large numbers, next year there will be less."

Where is this idea wrong? And what might the manager have actually meant?

Both are correct

$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})$	$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$	$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3$
25	2,03	0,0811
100	4,05	0,0405
400	8,29	0,0207
6400	31,81	0,0050
increases decreases		

Correct: Next year itself is expected to be around the mean (Regression to the Mean) — but this year's overpayment is not offset. It simply gets smaller within the overall portfolio size.

Calculated based on 2,000 independent replicates (left table).

Central Limit Theorem

Central Limit Theorem

MLT (Central Limit Theorem)

$X_1, \dots, X_n$ m.b.b. $\sim$, $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$ finite. Then

$$Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{\text{by distribution}} \mathcal{N}(0, 1),$$

regardless of the form of the prime set.

Three equally strong entries

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \sum X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

$\sim$ — “approximately distributed”. Not an exact equality.

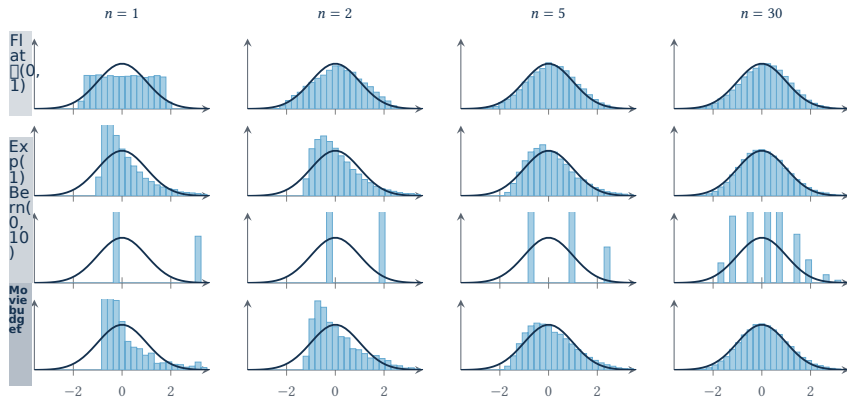
What MLT DOES NOT SAY “In a large sample, the data will be normal.” No — the data will remain as they are.

“The original set remains normal.” The original set does not change at all.

✗ “ $n \geq 30$ is a theorem.” Just a rough suggestion.

Only the sampling distribution of $\bar{X}$ approaches normal.

MLT in progress: four head sets, four $\square$



In each cell: 20,000 values of standardized $\square$ and density $\square(0, 1)$. All axes are on the same scale.

Common error

In $\text{Bern}(0, 10)$, the columns $\square \leq 2$ exceed the axis — this is not a defect: MLT is not yet working.

When is the “ $n \geq 30$ ” rule not sufficient

We measure convergence by the Kolmogorov Distance $\sup_x |\Phi_n(x) - \Phi(x)|$ — not by the assumption that “it looks normal to the eye.”

Prime set $n=1$ $n=2$ $n=5$ $n=30$ $n=100$

Flat	0.00	0.020	0.007	0.011	0.007	Movie
budget	1.62	0.112	0.052	0.020	0.012	Exp(1)
	2.00	0.101	0.064	0.033	0.025	Bern(0.10)
	2.67	0.493	0.366	0.146	0.082	

Look at the $n=30$ column: the error in the flat prime set is 0.011, and in Bern(0.10) it is 0.146 — 13 times larger. Same n , completely different quality.

The correct rule

The required n prime number depends on the skewness of the set:

$$\gamma_1(\bar{X}) = \frac{\gamma_1(X)}{\sqrt{n}}$$

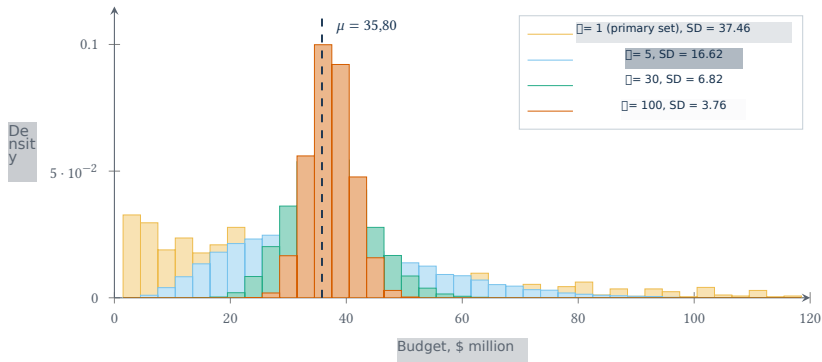
Movie budget: $n=1 = 1.617$. At $n=30$

$1.617/\sqrt{30} = 0.295 - 0.291$ in the simulation. ✓

A common mistake

Practical advice: Look at the skewness of the data instead of “ $n \geq 30$ ”. If it is strongly skewed, $n=30$ is less; if it is symmetric, $n=10$ is enough.

One prime set, as σ increases — on one axis



Two things happen at the same time. The distribution narrows (KSQ) and becomes symmetric (MLT). Both at $\sqrt{n}$ speed.

Audience question

Question: The curve at $\sigma = 1$ is the prime set itself. Why is it skewed to the right, but at $\sigma = 100$ it is almost symmetric? Have the budgets changed?

When MLT DOES NOT WORK: Cauchy Distribution

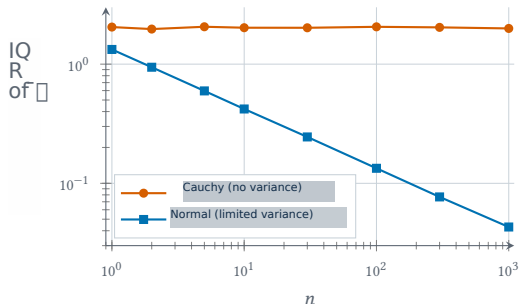
Condition violated Cauchy Distribution: density

$$f(x) = \frac{1}{\pi(1+x^2)}.$$

The tails are so “heavy” that $E[\bar{x}]$ does not exist — hence $\text{Var}(\bar{x})$ is also undefined (which would be infinite $E[\bar{x}^2]$). The main condition of MLT is not met.

Theorem The average of $\bar{x}$ Cauchy observations is exactly the same Cauchy distribution. Increasing n does not improve anything.

$P(|\bar{x}| > 10)$ remains ≈ 0.064 at every n ($n=1$: 0.0628; $n=1000$: 0.0667).



Audience Question

Blue line falls at $1/\sqrt{n}$ — MLT works. Red line is horizontal: 1 observation and 1,000 observations are equally accurate. Important in economics: financial returns, insurance losses, city sizes — heavy-tailed.

Selective distribution of shares

$\hat{p}$ — a special case of the mean

Trick: the proportion is also the mean

Let each observation be $X_i \in \{0, 1\}$ (1 is “success”). Then

$$\hat{p} = \frac{\text{successes}}{n} = \bar{X}.$$

For $\text{Bern}(p)$, $E[X_i] = p$, $\text{Var}(X_i) = p(1-p)$, so MLT is automatically applied:

$$E[\hat{p}] = p, \text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

Actual data: 2,467 movies

PG-13 proportion $p = 896/2467 = 0.3632$ (9 movies have no rating — they are “not PG-13”).

n	$\text{SD}(\hat{p})$	$\sqrt{\frac{p(1-p)}{n}}$	np
10	0,1524	0,1521	3,6
25	0,0961	0,0962	9,1
100	0,0477	0,0481	36,3

20,000 simulated samples. The formula is true for all n —but not normality.

Conditions Success-failure condition: to use the normal approximation, $np \geq 10$ and $n(1-p) \geq 10$. In the example above, this is true for $n \geq 28$. The condition applies only to the normal approximation, not to the formula.

Normal approximation to binomial and continuity correction

$X \sim \text{Bin}(50; 0.30)$. $\mu = 15$, $\sigma = \sqrt{50 \cdot 0.3 \cdot 0.7} = 3.2404$.

x	exact $P(X \leq x)$	normal	normal +1
9	0,0402	0,0320	0,0448
12	0,2229	0,1773	0,2202
15	0,5692	0,5000	0,5613
18	0,8594	0,8227	0,8600
21	0,9749	0,9680	0,9776

Error at $x = 12$: 0.0456 without correction, 0.0027 with correction — 17 times more accurate.

Continuity correction When approximating discrete X with a continuous normal, each integer is given a "column" of width ± 1

2:

$$P(X \leq x) \approx \Phi\left(\frac{x + 0.5 - \mu}{\sigma}\right)$$

A common mistake

Modern software can calculate the binomial exactly (`pbinom`, `binom.cdf`) — so the need for approximation is reduced.

Why do we study it? Because the confidence interval for p (week 9) and the Z -test (week 11) rely on this approximation.

Implementation and planning

Full problem — from beginning to end

Conditions Hypothetical model (constructed for estimation, not official statistics): wheat yield in the region $\mu = 4.50$ t/ha, $\sigma = 0.60$ t/ha. Randomly selected $n = 36$ farms.

Step 1 — check conditions. Farms are randomly selected $n = 36 \geq 30$ and the main set is not strongly skewed

Step 2 — SE. $SE = 0.60$
$$SE = \frac{0.60}{\sqrt{36}} = \frac{0.60}{6} = 0.10$$

Step 3 — standardization. $P(\bar{x} > 4.65)$:

$$z = \frac{4.65 - 4.50}{0.10} = +1.50$$

$$P = 1 - \Phi(1.50) = 0.0668$$

$$P(\bar{x} < 4.40): z = -1.00, \Phi(-1.00) = 0.1587 \quad P(4.40 < \bar{x} < 4.65) = 0.9332 - 0.1587 = 0.7745$$

$$\text{Central 95 \%: } 4.50 \pm 1.96 \times 0.10 = [4.304; 4.696]$$

For one household: $4.50 \pm 1.96 \times 0.60 = [3.324; 5.676]$

The interval is 6 times narrower — that's exactly $\sqrt{36} = 6$.

A common mistake

Note: It is wrong to read $z = 1.50$ as “the probability of getting 4.65 t/ha.” It is about the average of 36 farms. For one farm, $z = (4.65 - 4.50)/0.60 = 0.25$ and the probability is 0.401 — more than six times.

Sample size planning

Reverse the formula

$$\frac{SE}{\sqrt{n}} \leq E \implies n \geq \left(\frac{\sigma}{E}\right)^2$$

Movie budget, $\sigma = \$37.40$ million: target SE required E

\$10 million	14 5
million \$56	\$2
million 350	1
million \$1,400	

For the share — when σ is unknown

$\sigma(1 - \sigma)$ is the largest at $\sigma = 0.5$: 0.25. So we get the worst case: $\sigma \geq 0.25/E$.

target SE required E

0,05	100
0,03	278
0,02	625
0,01	2 500

Now you know why “ $\pm 3\%$ ” is standard in surveys: $E \approx 1,100$ (with the calculation of $1.96 \times SE$).

Common Error

Most important caveat. These formulas only control for random error. They do not reduce Sampling Bias at all: in a poorly designed questionnaire, $n = 100,000$ does not give a better answer than $n = 100$ —it only increases the confidence in the wrong answer. (Literary Digest, Week 1, 1936.)

**Summar
y**

Misconception Why it is wrong

“MLT normalizes the data.” The data does not change. Only the sampling distribution of $\bar{x}$ approaches normal. If you draw a histogram — it is still skewed. “MLT works if $n \geq 30$.” At $\text{Bern}(0.10)$ $n = 30$, the Kolmogorov error is 0.146 — unacceptable. The required n depends on the skewness. “SE is the standard deviation.” s is the dispersion of the observations, and SE is the mean. As n increases, SE decreases, but as s does not change. “A large sample is needed for a large population.” $\text{SE} = s/\sqrt{n}$ does not have n . If $p < 0.05$, p is insignificant. It is the same for China and Andijan. “The probability of a number coming after 10 consecutive numbers is higher.” The KSQ does not cover the past, it dilutes it. $|p - p/2|$ increases; only $|\hat{p} - 0.5|$ decreases. “I calculated $\hat{p}$ — this is p .” $\hat{p}$ — a realization of p . SE measures exactly “how far apart it can be.” “ p is large enough, the shift disappears.” p only reduces random error. Systematic shift is independent of p — it never disappears.

Main results of week 7

Must remember

$$E[\hat{p}] = p \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$SE(\hat{p}) = \sqrt{\text{Var}(\hat{p})} = \sqrt{\frac{p(1-p)}{n}}$$

$$E[\hat{p}] = p \quad \hat{p} - p \sim N(0, \frac{p(1-p)}{n})$$

Note: Independence is not required for $E[\hat{p}] = p$; it is required for $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$.

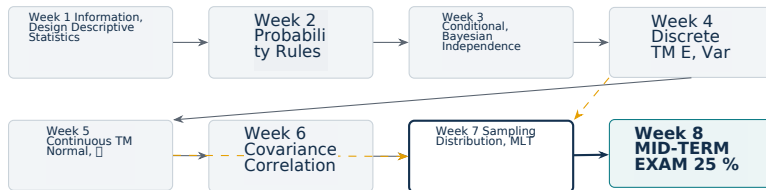
In three sentences

1. Statistics is a random variable; it has a distribution and we know it.
2. KSQ: $\hat{p}$ approaches. MLT: tells us what form the approximation will take.
3. $\sqrt{n}$ — the central cost of statistics: precision increases at the rate of the square root of the sample size, and the cost is linear.

Next step

So far we have taken p and q as known—that was a trick. In week 9, we will take them as unknown and construct a confidence interval. What happens when we substitute $\hat{p}$ for p ? $\rightarrow \hat{p}$ -distribution.

Week 8: Midterm Exam — Repetition Map



Exam Format: 2 hours. Section A — 15 short questions (40%). Section B — 3 problems (60%). Formula tables and calculators are allowed; phones are not allowed.

Audience Question

The yellow dotted lines are the most highly rated connections: Week 4 (E, Var) and Week 5 (χ^2 -rating) live within Week 7. Don't repeat them separately — repeat them together.

References

Main reading

- ▶ Çetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics (IMS, 2nd ed.), Chapters 12–13 — “Foundations for inference”. Open source: openintro.org/book/ims
- ▶ Newbold P., Carlson W., Thorne B. Statistics for Business and Economics (9th ed.), Chapter 6: “Distributions of Sample Statistics”.

Supplementary

- ▶ Freedman D., Pisani R., Purves R. Statistics (4th ed.), Chapters 16–18 — The best non-verbal explanation of MLT.
- ▶ Heumann C., Schomaker M., Shalabh. Introduction to Statistics and Data Analysis, Chapter 9.
- ▶ Tversky A., Kahneman D. (1971). Belief in the law of small numbers. Psychological Bulletin 76(2), 105–110. — a classic article on why people over-believe small samples.

Interactive

- ▶ seeingtheory.brown.edu — “Probability Distributions” and “Central Limit Theorem” sections.
- ▶ onlinestatbook.com/stat_sim/sampling_dist — Rice University’s classic MLT simulator.
- ▶ Reference: OWID/FAOSTAT (productivity), [movies-budget.csv](#).